

Advanced SQL NULL FUNCTIONS

Q1

SELECT employee-id, name, IFNULL(salary, 0) AS salary-with-default.

FROM Employees;

employee-id	name	salary-with-default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2

SELECT employee-id, name, IFNULL(department, 'Not Assigned') AS department-name

FROM Employees;

employee-id	name	department-name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Q3

SELECT order-id, customer-id

FROM Orders

WHERE ISNULL(customer-id);

order-id	customer-id
103	NULL

Q4

SELECT order-id, customer-id, IFNULL(DATE_FORMAT(delivery-date, 'yyyy-MM-dd'), 'Pending') AS delivery-status

FROM Orders;

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Q5

SELECT student_id, name, IFNULL(grade, '0') AS
final_grade
FROM Students

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6

SELECT COUNT(name) AS not_graded_count
FROM Students
WHERE grade IS NULL;

not_graded_count
1

Q7

SELECT product_id, name, price - IFNULL(discount, 0)
AS final_price
FROM Products;

product_id	name	final_price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Q8

```
SELECT COUNT(*) AS missing_email_count
FROM Customers
WHERE email IS NULL;
```

missing_email_count
2

Q9

```
SELECT customer_id, name, IFNULL(email, 'No email') AS
email_display
FROM Customers;
```

customer_id	name	email_display
1	Linda	No email
2	Joseph	joseph@gmail.com
3	Nia	No email

Q10

```
SELECT payment_id, IFNULL(method, 'Unknown') AS
method_display, status
FROM Payments;
```

payment_id	method_display	status
301	Credit	NULL
302	Pay Pal	Success
303	Unknown	Failed

Q11

```
SELECT item_id, item_name, IFNULL(quantity, 0) AS
quantity_checked
FROM Inventory;
```

item-id	item-name	quantity-checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Q12

SELECT emp-id, COALESCE (bonus, commission) AS
first-available-reward
FROM employees_extra;

emp-id	first-available-reward
1	300
2	100
3	NULL

Q13

SELECT COUNT(*) no-room-count
FROM classes
WHERE room IS NULL;

no-room-count
2

Q14

SELECT student-id, date, IFNULL(status, 'Not Marked')
AS attendance-status
FROM Attendance;

student-id	date	attendance-status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Q15

```
SELECT account-id, IFNULL(account-type, 'Unknown')
      AS type-display, IFNULL(balance, 0) AS
      balance-checked
```

FROM Bank-Accounts;

account-id	type-display	balance-checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Q16

```
SELECT project-id, title, IFNULL(DATE_FORMAT(start-date,
      'yyyy-MM-dd'), 'TBD') AS display-start-display
```

FROM Projects;

project-id	title	start-display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Q17

```
SELECT review-id, product-id, IFNULL(comment, 'No
      comment') AS comment-display, IFNULL(rating, 0)
      AS rating-display
```

FROM Reviews;

review-id	product-id	comment-display	rating-display
1	S01	Great product	4
2	S02	No comment	0
3	S03	Works fine	3

Q18

```
SELECT supplier_id, name, COALESCE (CAST (phone AS
    STRING, CAST (alt_phone AS STRING), 'No
    Contact') AS contact_number
```

FROM Suppliers;

supplier_id	name	contact_number
1	Global Goods	123 456 789
2	Best Suppliers	987 654 321
3	Value Source	No Contact

Q19

```
SELECT user_id, IFNULL(theme, 'Light') AS theme_set,
    IFNULL (language, 'English') AS language_set,
    IFNULL (timezone, 'UTC') AS timezone_set
```

FROM User_Settings;

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC + 1
3	Light	English	UTC

Q20

```
SELECT record_id, machine_id, IFNULL(issue, 'Unknown
    Issue') AS issue_log, IFNULL (technician, 'Not
    Assigned') AS technician_name
```

FROM Maintenance;

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex